

A guideline-based program may improve the outcome of stroke among illiterate patients

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Abstract

Background: Education level is considered as an important factor for post-stroke functional outcome. Our previous study confirms the feasibility of SMART program to improve adherence of secondary stroke prevention in China.

Aims: We aim to investigate whether this program can influence the impact of educational level on post-stroke functional outcome in this sub-group analysis of SMART study.

Methods: We enrolled 3722 patients with acute ischemic stroke from multicenter ($n=47$) SMART cohort. Patient's educational level was categorized as three: illiterate, 6th grade or lower, and 7th grade or higher. Functional outcome at six-month post-stroke was assessed by Modified Rankin Scale (mRS) and categorized as favorable (mRS: 0–2) or poor (mRS: 3–5). Binary logistic model, adjusting for age, gender, stroke history, hypertension, diabetes, hyperlipidemia, educational level, and NIHSS was performed, respectively, in SMART group and usual care group to evaluate the effect of the SMART program on stroke functional outcome among different educated patients.

Results: In both groups, logistic regression analysis showed that poor functional outcome was independently associated with older age [OR(95% CI): 1.81 (1.41–2.33), $p < 0.001$ (SMART); OR (95% CI): 1.38 (1.09–1.73), $p = 0.007$ (usual care)], higher baseline NIHSS [OR (95% CI): 2.90 (2.38–3.53), $p < 0.001$ (SMART); OR (95% CI): 2.82 (2.32–3.44), $p < 0.001$ (usual care)], and diabetes [OR(95% CI): 1.88 (1.18–3.00), $p = 0.008$ (SMART); OR (95% CI): 1.68 (1.03–2.73), $p = 0.037$ (usual care)]. Compared to illiterate, higher educational levels were independently associated with favorable outcome in usual care group [OR (95% CI): 0.43 (0.20–0.93), $p = 0.032$ (6th grade or lower) and OR (95% CI): 0.40 (0.19–0.84), $p = 0.016$ (7th Grade or higher)], but not in SMART group ($p = 0.806$ and $p = 0.889$, respectively).

Conclusions: This study demonstrates that implementation of SMART program may improve post-stroke functional outcome among illiterate. This suggests special attention should be paid to illiterate patients with intensive education in order to improve post-stroke disability.

Keywords

Stroke, functional outcome, education, guideline, SMART, program

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Introduction

Ischemic stroke is the leading cause of adult disability and the fourth cause of mortality in the USA.¹ About 55–75% of stroke survivors continue to experience motor deficits with poor quality of life.^{2,3} Infarct pattern, stroke subtype, initial NIHSS level, interventional strategies and age are well-known factors that predict functional outcome after stroke.^{4–8} In China, the leading cause of disability-adjusted life-years (DALYs) in 2010 was cardiovascular diseases (stroke and ischemic heart disease).⁹ The patients with ischemic stroke in China might have worse functional outcome than

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those in western countries.¹⁰ Given its possible impact on adherence to post-stroke treatment, educational level is considered as an important predicting factor for functional outcome after stroke, especially in developing countries.^{2,11,12} In recent decades, public health education efforts have been made towards not only to improve clinical and functional outcomes after stroke, but also to primary prevention of stroke. Evaluating the effect of educational level on the prognosis of stroke and thus developing more specific patient education strategies in lower educated patients are essential considering the fact that illiterate account for a large percentage of stroke population in China.

Our previous study confirms the feasibility of implementation of a structured guideline-based program (SMART program), comprising of pharmaceutical treatment, lifestyle modification and patient education, to improve adherence to the interventions of secondary stroke prevention in China.¹³ We thus hypothesize that this program may improve the adverse impact of lower education on functional outcome.

Materials and methods

Study participants

This sub-analysis was based on the SMART study, which was a nationwide, multi-center and cluster-randomized controlled trial to assess the effectiveness of a guideline-based structured care program for secondary stroke prevention, as opposed to usual care treatment, in China. Details of the SMART study have been published elsewhere.¹⁴ In brief, the SMART program comprised a combination of multiple interventions based on the current guidelines for the secondary prevention of ischemic stroke: pharmaceutical treatment targeting to specific pathological subtypes, lifestyle modification, and patient education. Physicians discussed lifestyle modification with the patients, including smoking cessation, healthy diet, and regular exercise. Patient education included an interactive education session emphasizing the importance of adhering to the SMART program. A total of 3821 participants were included in the SMART study, 2601 of whom were males (68.1%). We excluded 99 patients with diagnosis of transient ischemic attack but not stroke, leaving 3722 patients for this sub-group analysis. Of them, 1777 were assigned to receive SMART program and 1945 for usual care group.

The SMART study was approved by the central ethics committee at the leading study center at Peking Union Medical College Hospital and by the ethics committees at all participating sites. The written informed consent was obtained from all participants or their legal surrogates.

The definition of educational level and functional outcome

Patient's educational level was categorized as three: illiterate, 6th Grade or lower and 7th Grade or higher.¹⁵ Functional outcome at six-month post-stroke was assessed by Modified Rankin Scale (mRS) score and categorized as favorable (mRS: 0–2) or poor (mRS: 3–5). A total of 614 patients (16.5% of the whole cohort) were lost to follow up at six-month after index stroke. Finally, the mRS was available at six-month after stroke for 1445 of 1777 patients in SMART group and 1663 of 1945 patients in usual care group.

Definitions of cerebrovascular risk factors

Hypertension was defined as receiving antihypertensive drugs or blood pressure $\geq 140/90$ mmHg. Hyperlipidemia was defined as the use of lipid-lowering drugs or overnight fasting cholesterol level ≥ 200 mg/dl. Diabetes mellitus (DM) was defined as receiving antidiabetic drugs, or fasting blood sugar ≥ 126 mg/dl, or 2-h postprandial blood sugar ≥ 200 mg/dl. Smoking status was categorized as current smoker (current or quit smoking ≤ 6 months), ex-smoker (quit smoking > 6 months) or lifelong nonsmoker. History of ischemic stroke was defined as both self-report and medical records.

Statistical analysis

Statistical analysis was performed using SPSS version 19.0. Binary logistic regression model was respectively performed in SMART and usual care group. The association of education level on functional outcome was assessed in models adjusted for age, gender, history of hypertension, diabetes, hyperlipidemia and ischemic stroke, education, and baseline National Institutes of Health Stroke Scale (NIHSS). For all statistical test, $p < 0.05$ was considered statistically significant.

Results

The baseline demographic and clinical characteristics are given in Table 1. No significant difference for educational level was found between the groups ($p = 0.305$), although patients in SMART group were slight older ($p = 0.047$), more likely to have a history of hyperlipidemia ($p = 0.028$), and had higher NIHSS scores at baseline than those in usual care group ($p < 0.001$).

Binary logistic model, adjusting for age, gender, history of ischemic stroke, hypertension, diabetes and hyperlipidemia, patient's educational level, and baseline NIHSS, was respectively performed in SMART (Table 2) and usual care group (Table 3) to evaluate the potential predicting factors for 6-month post-stroke

Table 1. Demographics and baseline characteristics

Patients characteristics	SMART group (n = 1777)	Usual care group (n = 1945)
Male sex (%)	67	69.8
Age (SD) ^a	61.9 (11.6)	61.1(11.8)
Past medical history (%)		
Hypertension	64.9	63.8
Diabetes mellitus	23.6	21.2
Hyperlipidemia ^a	18.5	15.6
Previous ischemic stroke	24.5	24.2
Myocardial infarction	22.0	19.5
Atrial fibrillation	4.1	4.1
Smoking (%)		
Lifelong nonsmoker	54.3	47.8
Ex-smoker	15.4	17.4
Current smoking	30.3	34.8
Education level (%)		
Illiterate	7.0	6.6
6th Grade or lower	20.9	19.9
7th Grade or higher	68.4	72.6
Toast classification (%)		
Large artery	44.7	40.1
Small vessel disease	47.3	53.7
Cardioembolism	2.5	2.3
Other determined pathogenesis	0.5	0.4
Undetermined pathogenesis	4.9	3.7
Mean NIHSS at baseline (SD) ^a	4.9(4.4)	4.4(4.2)

^aThese characteristics were statistically significantly different between the two groups.

functional outcome. Older age (OR (95%CI): 1.81 (1.41–2.33), $p < 0.001$ in SMART group; OR (95% CI):1.38 (1.09–1.73), $p = 0.007$ in usual care group), higher baseline NIHSS (OR (95% CI): 2.90 (2.38–3.53), $p < 0.001$ in SMART group; OR (95% CI): 2.82 (2.32–3.44), $p < 0.001$ in usual care group), and

presence of DM (OR (95% CI):1.88(1.18–3.00), $p = 0.008$ in SMART group; OR (95% CI):1.68 (1.03–2.73), $p = 0.037$ in usual care group) were shown to be independent predictors for poor functional outcome at six months in both groups. Of note, compared to illiterate, higher educational levels were independently associated with favorable outcome in usual care [OR (95% CI): 0.43 (0.20–0.93), $p = 0.032$ (6th Grade or lower) and OR (95% CI): 0.40 (0.19–0.84), $p = 0.016$ (7th Grade or higher)], but not in SMART group ($p = 0.806$ and $p = 0.889$, respectively) (Tables 2 and 3).

Discussion

Our study demonstrates that implementation of the SMART program may potentially level off the adverse effect of illiterate on post-stroke functional outcome. These results suggest that more intensive patient education should be considered in lower educated patients to improve post-stroke disability.

Older Age, higher baseline NIHSS, and presence of DM were found to be independent predictors for poor functional outcome at six months in our patient cohort, which was consistent with previous findings observed from numerous studies focusing on risk factors of post-stroke functional outcome.^{4–6,16} Moreover, these associations were found not only in SMART group, but also in usual care group. These suggest that the SMART program has no effect on these preexisting risk factors of functional outcome which have already been considered in secondary prevention guideline of stroke.

In contrast, the independent and inverse effect of educational level on the functional outcome was only observed in usual care group, suggesting that lower educational level may have a negative impact on the functional outcome. This is in line with previous findings from limited studies.^{2,11,12} However, such effect was not observed in SMART group, which suggests that the patient education component in the SMART program may have leveled off such an effect, presumably by improving adherence to recommended stroke intervention approaches.

Educational level, as a marker of socioeconomic status, might influence the post-stroke outcome and should be considered in prevention and treatment strategies of stroke.¹⁷ The underlying mechanism of effect of educational level on functional outcome may include the following: patients with lower educational level might have less adherence to treatment of stroke including medication, rehabilitation practice, lifestyle and risk factors modification compared with those with higher educational level, either because of the subjective compliance or the objective economic burden/lack of access to health care facilities. This is consistent

Table 2. Potential risk factors to six-month post-stroke functional outcome in SMART group

	Crude distribution		mRS > 2 vs mRS ≤ 2	
	mRS ≤ 2 (n = 1296)	mRS > 2 (n = 149)	OR (95%CI)	p
Age, mean(SD)	61.33 (11.32)	67.27 (10.56)	1.81 (1.41–2.33)	<0.001
Male gender	68.0	59.7	0.928 (0.59–1.47)	0.751
Disease history				
Hypertension	64.0	71.7	1.27 (0.78–2.07)	0.332
Hypercholesterolemia	18.3	15.7	0.63 (0.35–1.16)	0.138
Diabetes mellitus	21.9	34.7	1.88 (1.18–3.00)	0.008
Previous Ischemic stroke	23	32.7	1.36 (0.85–2.18)	0.203
Education				
Illiterate	6.6	9.0	–	–
6th Grade or lower	20.1	26.9	1.11 (0.48–2.58)	0.806
7th Grade or higher	72.5	64.1	1.06 (0.47–2.38)	0.889
NIHSS, mean (SD)	4.32 (3.80)	9.84 (5.22)	2.90 (2.38–3.53)	<0.001

Note: All data are presented as percentage unless otherwise indicated. P values correspond to the relationship between each variable and the 6-month functional outcome after stroke. For continuous variables, the OR estimates the association related to an increase of 1 SD.

Table 3. Potential risk factors to six-month post-stroke functional outcome in usual care group

	Crude distribution		mRS > 2 vs mRS ≤ 2	
	mRS ≤ 2 (n = 1535)	mRS > 2 (n = 128)	OR (95% CI)	p
Age, mean(SD)	60.65(11.78)	65.25(11.03)	1.38 (1.09–1.73)	0.007
Male Gender	70.6	64.1	1.10 (0.66–1.82)	0.721
History of disease				
Hypertension	63.0	72.4	1.32 (0.83–2.11)	0.246
Hypercholesterolemia	15.5	18.6	1.33 (0.74–2.38)	0.340
Diabetes mellitus	19.8	29.7	1.68 (1.03–2.73)	0.037
Previous Ischemic stroke	23.9	30.7	1.08 (0.66–1.78)	0.759
Education				
Illiterate	6.1	15.2	–	–
6th Grade or lower	20.8	24.8	0.43 (0.20–0.93)	0.032
7th Grade or higher	73.1	60.0	0.40 (0.19–0.84)	0.016
NIHSS, mean (SD)	4.02 (3.78)	9.41 (5.56)	2.82 (2.32–3.44)	<0.001

Note: All data are presented as percentage unless otherwise indicated. P values correspond to the relationship between each variable and the 6-month functional outcome after stroke. For continuous variables, the OR estimates the association related to an increase of 1 SD.

with the fact that much higher prevalence of stroke was observed in rural than in the urban area of China.^{18,19} Up to now, only a few studies have explored the effects of educational level on the functional outcome after stroke in developed countries.^{2,12} This might be due to the obvious difference of the averaged educational level between developed countries and developing countries. Howard et al. report that educational level did not influence the probability of returning to work in 1980s, which might be due to the higher average educational level in USA.²⁰ As previously published by our team in this study, this SMART program comprises a combination of guideline-based active pharmaceutical treatment, lifestyle modification, and patient education. Of them, patient education may play an important role especially in those patients with lower educational level by improving their adherence to treatment, while those patients with higher educational level may have much better adherence to treatment after stroke to improve disability even without intensive patient education. As we know, illiterate account for a quite large proportion of stroke population in China. Compared with those with higher educational level, illiterate patients in general have poor adherence to stroke treatment strategies and thus have relative poor outcome. Further studies are needed to better understand how educational and income-related factors affect the functional outcome after stroke, which is important in health care provision.

The strengths of this present study included the prospective design and the large sample size. These results can serve as strong evidence for future policy making to improve the quality of stroke prevention in China. Our study also has a few potential limitations. First, there are some differences for the baseline demographic and clinical characteristics between the SMART and usual care group, which may have contributed to a selection bias. However, no significant difference for percentages of educational levels was found between the two groups. Second, the percentage of missing follow-up data was considerable. However, after adjustment of age and gender, no difference between the patients with and without follow-up information was observed for the baseline demographic and clinical characteristics, not only in the whole cohort but in SMART and usual care group separately, which makes the bias much less (data shown in the supplemental Tables 1 and 2). Third, pre-stroke mRS was not adjusted in the logistic model, which might influence the analysis result. However, the NIHSS is actually considered as an equilibrated tool for evaluating multiple aspects of the neurological status. Baseline NIHSS could largely capture the severity of pre-stroke disability. Moreover, the results remain unchanged even after further adjustment on pre-stroke mRS (data not shown). Last, educational level was unfortunately not listed in the baseline

characteristics in the main paper, even though it would not influence the result.

Summary

In conclusion, we found that implementation of SMART program may play an important role in improving post-stroke functional outcome among illiterate patients in China. Our findings reinforce that intensive educating strategies should be considered particularly in lower educated patients to improve post-stroke disability, especially in developing countries.

Authors' contributions

JN and MY contributed equally to this article.

Declaration of conflicting interests

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